



MATHEMATICS (EXTENSION 1)

2012 HSC Course Assessment Task 1

November 25, 2011

General instructions

- Working time – 50 minutes.
- Commence each new question on a new page. Write on both sides of the paper.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

NAME:

Class (please ✓)

- ☐ 12M3C – Mr Lowe
- ☐ 12M3D – Mr Berry
- ☐ 12M3E – Mr Lam
- ☐ 12M4A – Mr Fletcher
- ☐ 12M4B – Mr Ireland
- ☐ 12M4C – Mr Weiss

BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{9}$	$\overline{9}$	$\overline{7}$	$\overline{9}$	$\overline{15}$	$\overline{48}$	

Question 1 (9 Marks)	Commence a NEW page.	Marks
(a) Find the vertex and focus of $(x - 4)^2 = 2y - 6$.		3
(b) Find the equation of the locus of a point $P(x, y)$ which moves such that its distance from the line $x = 8$ is twice its distance from the point $(2, 0)$.		3
(c) A parabola has its focus at $(2, -4)$ and its directrix is the x axis.		
i. Determine its vertex and write down the equation of the parabola.		2
ii. What is the length of the latus rectum of the parabola?		1

Question 2 (9 Marks)	Commence a NEW page.	Marks
(a) If α , β and γ are roots of the equation $2x^3 - 5x + 3 = 0$, find the value of		
i. $\alpha\beta + \alpha\gamma + \beta\gamma$		1
ii. $\alpha\beta\gamma$		1
iii. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$		2
iv. $\frac{1}{\alpha^2 + \beta^2 + \gamma^2}$		2
(b) The polynomial $x^3 + 2x^2 + ax + b$ has a factor of $(x + 2)$ and when divided by $(x - 2)$ leaves a remainder of 12. Find the value of a and b .		3

Question 3 (7 Marks)	Commence a NEW page.	Marks
(a) Write down the equation of the chord of contact of the parabola $x^2 = 8y$ from the external point $P(4, -6)$. Do <i>not</i> derive the equation.		2
(b) Given the quadratic equation $x^2 + (m - 6)x - 8m = 0$		
i. Find the value of m if the roots are reciprocal of one another.		2
ii. For what values of m does the quadratic $x^2 + (m - 6)x - 8m = 0$ have real roots?		3

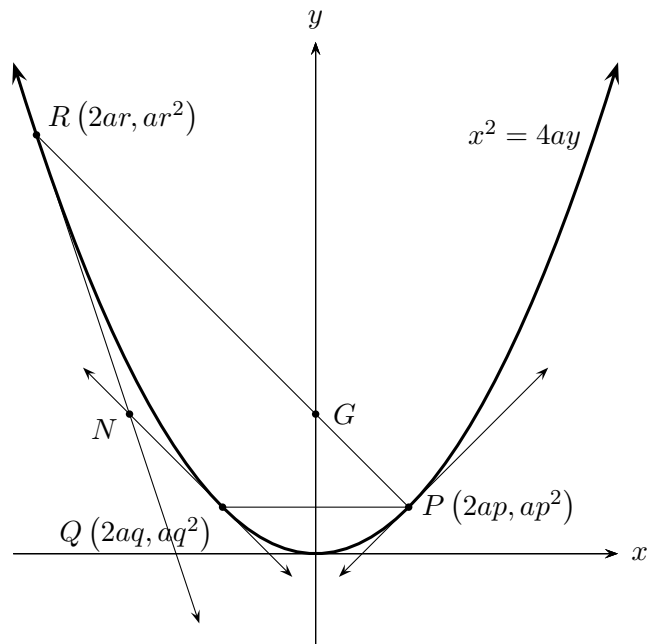
Question 4 (9 Marks)	Commence a NEW page.	Marks
(a) i. Show that $(x - 1)$ is a factor of $P(x) = x^3 - 6x^2 + 11x - 6$.		1
ii. Express $P(x)$ as a product of its factors.		3
iii. Without using calculus, solve the inequality $x^4 - 6x^3 + 11x^2 - 6x \leq 0$. <i>Hint:</i> a sketch may be useful.		2
(b) Solve $(x + x^{-1})^2 - 6(x + x^{-1}) + 8 = 0$.		3

Question 5 (15 Marks)

Commence a NEW page.

Marks

$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$.
 PQ is a chord perpendicular to the axis of symmetry of the parabola.



- (a) Derive the equation of the tangent to the parabola at point P . **2**

- (b) Prove that the equation of the normal at P is $x + py = 2ap + ap^3$. **2**

The normal at P cuts the axis of symmetry at G and the parabola again at R .

- (c) Find the coordinates of G . **2**

- (d) Find the locus of point G as P moves along the parabola. **1**

- (e) The tangents at Q and R meet at N . Show that N has coordinates **3**

$$(a(q + r), aqr)$$

- (f) Prove that $r = -p - \frac{2}{p}$. **2**

- (g) Prove that NG is perpendicular to the axis of symmetry. **2**

End of paper.

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Suggested Solutions

Question 1 (Fletcher)

(a) (3 marks)

- ✓ [1] for correct focal length a .
- ✓ [1] for vertex.
- ✓ [1] for focus.

$$(x-4)^2 = 2y-6$$

$$(x-4)^2 = 2(y-3) = 4 \times \frac{1}{2}(y-3)$$

$$\therefore a = \frac{1}{2} \quad V(4, 3) \quad S\left(4, \frac{7}{2}\right)$$

(b) (3 marks)

- ✓ [1] for correct distance formulae.
- ✓ [1] for correct equation.
- ✓ [1] for correct locus.

Let the point $P(x, y)$ be the variable point that moves.

- The distance from P to $x = 8$ is

$$d_1 = \sqrt{(x-8)^2}$$

- The distance from P to $(2, 0)$ is

$$d_2 = \sqrt{(x-2)^2 + y^2}$$

As P moves s.t. the distance from $(2, 0)$ is twice that to $x = 8$,

$$\sqrt{(x-8)^2} = 2\sqrt{(x-2)^2 + y^2}$$

$$(x-8)^2 = 4(x^2 - 4x + 4 + y^2)^2$$

$$\therefore \underset{-x^2}{x^2} - \underset{-16}{16x} + \underset{-16}{64} = \underset{-x^2}{4x^2} - \underset{-16}{16x} + \underset{-16}{16} + 4y^2$$

$$3x^2 + y^2 = 48$$

(c) i. (2 marks)

- ✓ [1] for vertex.
- ✓ [1] for equation.

$$V(2, -2)$$

$$(x-2)^2 = -4(2)(y+2)$$

$$(x-2)^2 = -8(y+2)$$

ii. (1 mark)

$$\text{Length of latus rectum} = 4a = 4 \times 2 = 8$$

Question 2 (Ireland)

(a) $2x^3 + 0x^2 - 5x + 3 = 0$.Note that $\alpha + \beta + \gamma = -\frac{b}{a} = 0$.

i. (1 mark)

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -\frac{5}{2}$$

ii. (1 mark)

$$\alpha\beta\gamma = -\frac{d}{a} = -\frac{3}{2}$$

iii. (2 marks)

- ✓ [1] for correct expression.
- ✓ [1] for final answer.

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{-\frac{5}{2}}{-\frac{3}{2}} = \frac{5}{3}$$

iv. (2 marks)

- ✓ [1] for correct expression.
- ✓ [1] for final answer.

$$\frac{1}{\alpha^2 + \beta^2 + \gamma^2}$$

$$= \frac{1}{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)}$$

$$= \frac{1}{0 - 2\left(-\frac{5}{2}\right)} = \frac{1}{5}$$

(b) ✓ [1] for two equations in a and b .✓ [1] each per correct value of a and b .As $(x+2)$ is a factor of

$$P(x) = x^3 + 2x - ax + b,$$

$$P(-2) = (-2)^3 + 2(-2)^2 - 2a + b = 0$$

$$-8 + 8 - 2a + b = 0$$

$$\therefore b = 2a \quad (1)$$

By the remainder theorem,

$$P(2) = (2)^3 + 2(2)^2 + 2a + b = 12$$

$$2a + b = -4 \quad (2)$$

Substitute (1) to (2),

$$2a + b = b + b = -4$$

$$b = -2$$

$$\therefore a = -1$$

Question 3 (Berry)

(a) (2 marks)

- ✓ [1] for correct equation of the chord
 $xx_0 = 2a(y + y_0)$.
 ✓ [1] for final answer.

$$x^2 = 8y = 4 \times 2y$$

$$\therefore a = 2$$

The equation of the chord of contact is

$$xx_0 = 2a(y + y_0)$$

From $(4, -6)$ with focal length $a = 2$,

$$4x = 4(y - 6)$$

$$x = y - 6$$

$$\therefore y = x + 6$$

(b) i. (2 marks)

- ✓ [1] for $\alpha\beta = \frac{c}{a} = 1$.
 ✓ [1] for correct value of m .

$$x^2 + (m - 6)x - 8m = 0$$

Roots are α & $\frac{1}{\alpha}$.

$$\alpha \times \frac{1}{\alpha} = \frac{c}{a}$$

$$\frac{-8m}{1} = 1$$

$$\therefore m = -\frac{1}{8}$$

ii. (3 marks)

- ✓ [1] for $\Delta = m^2 + 20m + 36$.
 ✓ [1] for identifying that $\Delta \geq 0$.
 ✓ [1] for final answer.

$$\Delta = b^2 - 4ac$$

$$= (m - 6)^2 - 4(1)(-8m)$$

$$= m^2 - 12m + 36 + 32m$$

$$= m^2 + 20m + 36$$

$$= (m + 18)(m + 2)$$

Real roots occur when $\Delta \geq 0$:

$$(m + 18)(m + 2) \geq 0$$

$$\therefore m \leq -18 \quad \text{or} \quad m \geq -2$$

Question 4 (Lam)(a) $P(x) = x^3 - 6x^2 + 11x - 6$

i. (1 mark)

$$P(1) = 1 - 6 + 11 - 6 = 0$$

By the factor theorem, $x - 1$ is a factor of $P(x)$.

ii. (3 marks)

- ✓ [3] for correctly factorising.
 ✓ [2] only for $(x - 1)(x - 6)(x + 1)$.
 ✓ [1] for one further error, depending on solution provided.

• By long division,

$$\begin{array}{r} x^2 - 5x + 6 \\ x - 1 \overline{) x^3 - 6x^2 + 11x - 6} \\ \underline{-x^3 + x^2} \\ -5x^2 + 11x \\ \underline{5x^2 - 5x} \\ 6x - 6 \\ \underline{-6x + 6} \\ 0 \end{array}$$

$$\therefore P(x) = (x - 1)(x^2 - 5x + 6)$$

$$= (x - 1)(x - 2)(x - 3)$$

• By guessing $x - 2$ is a factor:

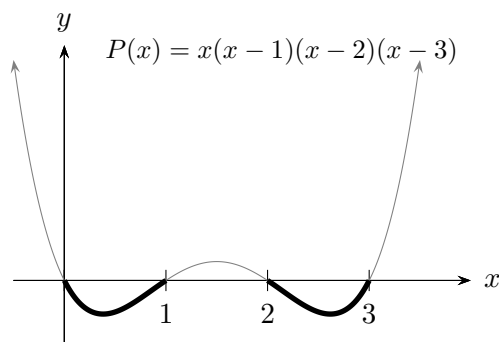
$$P(2) = 2^3 - 6(2)^2 + 11(2) - 6$$

$$= 8 - 24 + 22 - 6 = 0$$

Similarly, use factor theorem again for $x - 3$ to show it is also a factor.

iii. (2 marks)

- ✓ [1] for correct method (usually correct graph).
 If any other graph is (correctly) sketched and resulting in incorrect inequalities (i.e. $P(x) \leq 0$), only [1] is awarded. [0] for incorrect graph and incorrect inequalities (2 errors).
 ✓ [1] for evaluating $P(x) \leq 0$ correctly (i.e. correct inequality)



From the graph, $P(x) \leq 0$ when

$$0 \leq x \leq 1 \quad \text{or} \quad 2 \leq x \leq 3$$

Evaluating $x + \frac{1}{x} = 4$,

$$\begin{aligned} x + \frac{1}{x} &= 4 \\ \underbrace{x + \frac{1}{x}}_{\times x} &= 4 \times x \\ x^2 + 1 &= 4x \\ x^2 - 4x + 1 &= 0 \\ x &= \frac{4 \pm \sqrt{4^2 - 4(1)(1)}}{2} \\ &= \frac{4 \pm \sqrt{12}}{2} = \frac{\cancel{2}(2 \pm \sqrt{3})}{\cancel{2}} \\ &= 2 \pm \sqrt{3} \end{aligned}$$

Question 5 (Lowe/Weiss)

(b) (3 marks)

- ✓ [1] for method which leads to correct solutions.
- ✓ [1] for $x = 1$.
- ✓ [1] for $x = 2 \pm \sqrt{3}$.
- ✓ [-1] for non exact values.

$$\left(x + \frac{1}{x}\right)^2 - 6\left(x + \frac{1}{x}\right) + 8 = 0$$

Let $m = \left(x + \frac{1}{x}\right)$,

$$\begin{aligned} m^2 - 6m + 8 &= 0 \\ (m - 4)(m - 2) &= 0 \\ \therefore m &= 2, 4 \\ \therefore \left(x + \frac{1}{x}\right) &= 2, 4 \end{aligned}$$

Evaluating $x + \frac{1}{x} = 2$,

$$\begin{aligned} x + \frac{1}{x} &= 2 \\ \underbrace{x + \frac{1}{x}}_{\times x} &= 2 \times x \\ x^2 + 1 &= 2x \\ x^2 - 2x + 1 &= 0 \\ (x - 1)^2 &= 0 \\ \therefore x &= 1 \end{aligned}$$

(a) (2 marks)

- ✓ [1] for correct gradient.
- ✓ [1] for correct equation.

$$\begin{aligned} y &= \frac{x^2}{4a} \\ \frac{dy}{dx} &= \frac{2x}{4a} = \frac{x}{2a} \Big|_{x=2ap} = \frac{\cancel{4}ap}{\cancel{4}a} = p \end{aligned}$$

Apply the point gradient formula to find the equation of the tangent,

$$\begin{aligned} \frac{y - ap^2}{x - 2ap} &= p \\ y - \underset{+ap^2}{ap^2} &= \underset{+ap^2}{px - 2ap^2} \\ y &= px - ap^2 \end{aligned}$$

(b) (2 marks)

- ✓ [1] for correct gradient.
- ✓ [1] for correct equation.

$$m_{\perp} = -\frac{1}{p}$$

Apply the point gradient formula to find the equation of the normal,

$$\begin{aligned} \frac{y - ap^2}{x - 2ap} &= -\frac{1}{p} \\ y - ap^2 &= -\frac{1}{p}(x - 2ap) \\ py - ap^3 &= -x + 2ap \\ \therefore x + py &= 2ap + ap^3 \end{aligned}$$

(c) (2 marks)

- ✓ [1] for correct point G .
 ✓ [1] for correct locus equation.

At $x = 0$, the equation of the normal becomes

$$\begin{aligned} py &= 2ap + ap^3 \\ \therefore y &= 2a + ap^2 \\ \therefore G(0, 2a + ap^2) \end{aligned}$$

(d) (1 mark)

The locus of G is $x = 0$ (y axis)

(e) (3 marks)

- ✓ [1] for evaluating simultaneous equations involving tangents at Q and R .
 ✓ [1] for finding the x value.
 ✓ [1] for finding the y value.

Equations of tangents at Q and R are

$$\begin{cases} y = qx - aq^2 & \textcircled{1} \\ y = rx - ar^2 & \textcircled{2} \end{cases}$$

Equating $\textcircled{1}$ and $\textcircled{2}$ to find the point of intersection N ,

$$\begin{aligned} qx - aq^2 &= rx - ar^2 \\ qx - rx &= aq^2 - ar^2 \\ x(q - r) &= a(q^2 - r^2) = a(q - r)(q + r) \\ \therefore x &= a(q + r) & \textcircled{3} \end{aligned}$$

Substitute $\textcircled{3}$ to $\textcircled{1}$ to find the y coordinate:

$$\begin{aligned} y &= q(a(q + r) - aq^2) \\ &= \cancel{aq^2} + aqr - \cancel{aq^2} \\ &= aqr \\ \therefore N &\left(a(q + r), aqr\right) \end{aligned}$$

(f) (2 marks)

- ✓ [1] for $2ar - 2ap = ap^3 - ar^2p$.
 ✓ [1] for completely showing $r = -p - \frac{2}{p}$.

As $R(2ar, ar^2)$ lies on the normal from P , substitute its coordinates into the equation

of the normal:

$$\begin{aligned} x + py \Big|_{\substack{x=2ar \\ y=ar^2}} &= 2ap + ap^3 \\ 2ar + apr^2 &= 2ap + ap^3 \\ 2ar - 2ap &= ap^3 - apr^2 \\ 2\cancel{p}(r - p) &= \cancel{p}(p^2 - r^2) \\ \frac{-2(\cancel{p-r})}{\cancel{p}} &= \frac{p(\cancel{p-r})}{\cancel{p}}(p + r) \\ \frac{-2}{p} &= p + r \\ \therefore r &= -p - \frac{2}{p} \end{aligned}$$

Alternatively, obtain quadratic in r and solve as quadratic equation.

$$\begin{aligned} \cancel{p}pr^2 + 2\cancel{p}r - 2\cancel{p}p - \cancel{p}p^3 &= 0 \\ a = p \quad b = 2 \quad c &= -(2p + p^3) \\ r &= \frac{-2 \pm \sqrt{(2)^2 + 4(p)(2p + p^3)}}{2p} \\ &= \frac{-2 \pm \sqrt{4 + 8p^2 + 4p^4}}{2p} \\ &= \frac{-2 \pm \sqrt{4(p^2 + 1)^2}}{2p} \\ &= \frac{-2 \pm 2(p^2 + 1)}{2p} = \frac{-2 \pm 2p^2 + 2}{2p} \\ r &= \frac{-2 + 2p^2 + 2}{2p} \quad \Bigg| \quad r = \frac{-2 - 2p^2 - 2}{2p} \\ &= \frac{2p^2}{2p} = p \quad \Bigg| \quad = \frac{-4 - 2p^2}{2p} = -p - \frac{2}{p} \end{aligned}$$

As $r \neq p$, then $r = -p - \frac{2}{p}$ only.

(g) (2 marks)

- ✓ [1] noting $p = -q$ and substitutes this into coordinates of N .
 ✓ [1] for correct working.
 Since PQ is $\perp$ to y axis, $\therefore p = -q$. Also, $r = -p - \frac{2}{p}$.

$$\begin{aligned} y &= aqr = a(-p) \left(-p - \frac{2}{p}\right) \\ &= ap \left(p + \frac{2}{p}\right) \\ &= ap^2 + 2a \\ \therefore N &\left(a(q + r), 2a + ap^2\right) \end{aligned}$$

As $G(0, 2a + ap^2)$ and $N(a(q + r), 2a + ap^2)$, hence NG is perpendicular to the axis of the parabola.